Practical no. 19

Aim - Design and implement an embedded access-control prototype that grants entry only when either a whitelisted RFID tag is detected or a legitimate fingerprint is verified, provides real-time visual feedback on an LCD, indicates status via an LED, and pulses a relay (active-LOW) to unlock for a fixed interval. The system must log all events (time, method, result) for auditability.

Hardware Requirements

1. Raspberry Pi Model A/B/B+
2. Jumper wires (Female to Male)
3. Relay Module
4. RFID RC522
5. R307/R305 fingerprint sensor
6. 16×2 LCD

- **Software Requirements**
  1. Raspbian Stretch OS

1. **Connect with Raspberry Pi's GPIO Pins.**

- **Relay Module**

Device / Pin	Raspberry Pi 4	Physical Pin
Relay VCC	5V	2
Relay GND	GND	9
Relay IN	GPIO23	16

- **LCD 16X2 Connections**

Device / Pin	Raspberry Pi 4	Physical Pin
LCD GND	GND	6
LCD VCC	5V	4
LCD SDA	GPIO2 / SDA	3
LCD SCL	GPIO3 / SCL	5

- **R307/R305 Fingerprint**

VCC	5V	Pin 2
GND	GND	Pin 20
TX	GPIO15 / RXD	Pin 10
RX	GPIO14 / TXD	Pin 8

- **RC522 RFID → Raspberry Pi 4**

RC522 Pin	Raspberry Pi Physical Pin	GPIO
3.3V	Pin 1	3.3V
RST	Pin22	GPIO25
GND	Pin 30	GND
MISO	Pin 21	GPIO9
MOSI	Pin 19	GPIO10
SCK	Pin 23	GPIO11
SDA/SS	Pin 24	GPIO8

LED	Raspberry Pi	Physical Pin
LED anode (+)	GPIO21	40
LED cathode (-)	GND through resistor	e.g. Pin 34

Software Installation -

sudo apt update
sudo apt upgrade -y
sudo apt install -y python3-pip python3-serial python3-rpi.gpio i2c-tools
pip3 install --break-system-packages RPLCD
pip3 install --break-system-packages adafruit-circuitpython-fingerprint
pip3 install --break-system-packages mfrc522
sudo raspi-config
sudo i2cdetect -y 1

Install Python modules using --break-system-packages (required for Raspberry Pi OS Bookworm)
pip3 install RPLCD adafruit-circuitpython-fingerprint mfrc522 --break-system-packages

I2C LCD Address

The LCD shown in the supplied image uses an I2C backpack. The common address is **0x27**. If i2cdetect shows
0x3F, change the LCD address in the program from 0x27 to 0x3F.

CODE:-

Programme Number 01 (RFID)

```
from mfrc522 import SimpleMFRC522
import RPi.GPIO as GPIO
reader = SimpleMFRC522()
try:
    print("Place RFID card...")
    while True:
        uid, text = reader.read()
        print("RFID UID:", uid)
        print("Data:", text)
        break
finally:
    GPIO.cleanup()
```

Programme Number 02 (FINGERPRINT)

```
import serial
import adafruit_fingerprint
# UART configuration on Raspberry Pi GPIO 14/15
uart = serial.Serial("/dev/serial0", baudrate=57600, timeout=1)
```

```
finger = adafruit_fingerprint.Adafruit_Fingerprint(uart)
location = int(input("Enter fingerprint ID (1-127): "))
print("Place finger...")
while finger.get_image() != adafruit_fingerprint.OK:
    pass

if finger.image_2_tz(1) != adafruit_fingerprint.OK:
    print("Image conversion failed")
    exit()
print("Remove finger")
input("Press Enter after removing finger...")
print("Place same finger again...")
while finger.get_image() != adafruit_fingerprint.OK:
    pass

if finger.image_2_tz(2) != adafruit_fingerprint.OK:
    print("Image conversion failed")
    exit()

if finger.create_model() != adafruit_fingerprint.OK:
    print("Fingerprint mismatch")
    exit()

if finger.store_model(location) == adafruit_fingerprint.OK:
    print("Fingerprint enrolled successfully")
else:
    print("Enrollment failed")
```

Programme Number 03 (main Programme)

```
import RPi.GPIO as GPIO
import time
import csv
import os
import serial
from datetime import datetime
from RPLCD.i2c import CharLCD
from mfrc522 import SimpleMFRC522
import adafruit_fingerprint
# Pin Definitions (BCM numbering)
RELAY = 23
LED = 21
GPIO.setmode(GPIO.BCM)
GPIO.setup(RELAY, GPIO.OUT)
GPIO.setup(LED, GPIO.OUT)
# Active-LOW Relay: default HIGH (OFF)
GPIO.output(RELAY, GPIO.HIGH)
GPIO.output(LED, GPIO.LOW)
# Initialize I2C LCD (change 0x27 to 0x3F if i2cdetect reported 0x3F)
lcd = CharLCD(
    i2c_expander='PCF8574',
    address=0x27,
    port=1,
```

```
    cols=16,
    rows=2,
    charmap='A00',
    auto_linebreaks=True
)

reader = SimpleMFRC522()
# Replace with the decimal UID obtained from rfid_test.py
WHITELIST = [
    123456789,
    987654321
]
# Initialize Fingerprint Sensor
uart = serial.Serial("/dev/serial0", baudrate=57600, timeout=1)
finger = adafruit_fingerprint.Adafruit_Fingerprint(uart)
LOG_FILE = "access_log.csv"
def log_event(method, result):
    new_file = not os.path.exists(LOG_FILE)
    with open(LOG_FILE, "a", newline="") as file:
        writer = csv.writer(file)
        if new_file:
            writer.writerow(["Time", "Method", "Result"])
        writer.writerow([
            datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
            method,
            result
        ])
def show_message(line1, line2=""):
    lcd.clear()
    lcd.write_string(line1[:16])
    if line2:
        lcd.cursor_pos = (1, 0)
        lcd.write_string(line2[:16])
def unlock(method):
    show_message("ACCESS GRANTED", method)
    GPIO.output(LED, GPIO.HIGH)
    GPIO.output(RELAY, GPIO.LOW)  # Active-LOW trigger
    log_event(method, "GRANTED")
    time.sleep(5)
    GPIO.output(RELAY, GPIO.HIGH) # Turn off relay
    GPIO.output(LED, GPIO.LOW)
def deny(method):
    show_message("ACCESS DENIED", method)
    GPIO.output(LED, GPIO.LOW)
    log_event(method, "DENIED")
    time.sleep(2)
def check_fingerprint():
    # Poll sensor for up to 3 seconds
    start = time.time()
    while time.time() - start < 3:
        result = finger.get_image()
        if result == adafruit_fingerprint.OK:
            break
        elif result == adafruit_fingerprint.NOFINGER:
```

```
            time.sleep(0.1)
            continue
        else:
            return False
    else:
        return False

    show_message("Processing...")
    if finger.image_2_tz(1) != adafruit_fingerprint.OK:
        return False
    if finger.finger_search() != adafruit_fingerprint.OK:
        return False
    return True

try:
    show_message("ACCESS CONTROL", "SYSTEM READY")
    time.sleep(2)

    while True:
        show_message("Scan RFID", "or Fingerprint")
        # Non-blocking RFID check
        card_id, text = reader.read_no_block()
        if card_id:
            print("RFID UID:", card_id)
            if card_id in WHITELIST:
                unlock("RFID")
            else:
                deny("RFID")
            time.sleep(1)
            continue

        # Check Fingerprint sensor
        if check_fingerprint():
            print("Fingerprint matched")
            unlock("FINGERPRINT")
            time.sleep(1)
except KeyboardInterrupt:
    print("\nProgram stopped")
finally:
    GPIO.output(RELAY, GPIO.HIGH)
    GPIO.output(LED, GPIO.LOW)
    GPIO.cleanup()
    lcd.clear()
```

Running the Programs

python3 rfid_test.py

python3 enroll.py

python3 access_control.py